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Forecasting macroeconomic tail risk in real time: Do textual data add value?☆

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ABSTRACT

We examine the incremental value of news-based data relative to the FRED-MD economic indicators for quantile predictions of employment, output, inflation, and consumer sentiment in a high-dimensional setting. Our results suggest that news data contain valuable information not captured by a large set of economic indicators. We provide empirical evidence that this information can be exploited to improve tail risk predictions. The added value is largest when media coverage and sentiment are combined to compute text-based predictors. Methods that capture quantile-specific non-linearities produce superior forecasts to those with linear predictive relationships. The results are robust along different modeling choices.

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1. Introduction

Periods of economic stress, such as the Global Financial Crisis or the COVID-19 pandemic, have highlighted the importance of tail risk predictions. Quantile predictions of macroeconomic time series provide a more nuanced picture than point forecasts, since the former allows the predictive relationship between the target variable and the covariates to vary across quantiles. As policymakers and central banks are particularly interested in the tails associated with high economic risk, the literature on macroeconomic forecasting has paid increasing attention to quantile predictions (see, e.g., Adams, Adrian, Boyarchenko, & Giannone, 2021; Adrian, Boyarchenko, & Giannone, 2019; Carriero, Clark, & Marcellino, 2022a; Clark, Huber, Koop, Marcellino, & Pfarrhofer, 2023; Korobilis, 2017; Manzan, 2015; Prüser & Huber, 2024).

Another recent development in macroeconomic forecasting is the use of textual data, which provides timely information that may embed complementary signals to (hard) economic indicators (see, e.g., Bybee, Kelly, Manela, & Xiu, 2024; Ellingsen, Larsen, & Thorsrud, 2022; Kalamara, Turrell, Redl, Kapetanios, & Kapadia, 2022; Larsen & Thorsrud, 2019; Shapiro, Sudhof, & Wilson, 2022; van Dijk & de Winter, 2023). Researchers seek to quantify the predictive relationship between narratives and economic outcomes by extracting information from textual data. As argued by Shiller (2017), narratives may even have the power to shape economic outcomes causally. Therefore,

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quantifying changing narratives based on textual data seems to be a worthwhile endeavor.

Our study takes a data-driven approach to assess whether textual data, relative to a large set of economic indicators, add value to macroeconomic quantile predictions in high-dimensional settings. We focus on monthly tail risk forecasts (nowcasts and one month ahead) of employment, industrial production, inflation, and consumer sentiment between 1999:08 and 2021:12. To distinguish between the benefits of text-based predictors for tail predictions compared to the center of the distribution, we look at a broad range of quantiles (5%, 10%, 25%, 50%, 75%, 90%, 95%). As data revisions are an important issue in out-of-sample (OOS) macroeconomic forecasting, we use real-time vintages of economic predictors contained in the prominent McCracken and Ng (2016) FRED-MD database.

For the computation of text-based predictors, we opt for an unsupervised machine learning approach that is agnostic about the dominant themes in a corpus of documents, namely the Correlated Topic Model (CTM) by Blei and Lafferty (2007). We use the CTM to map nearly 800,000 newspaper articles from *The New York Times* and *The Washington Post* into a set of numerical text-based predictors. The most frequent words within each topic (word cluster) characterize the topic, and the proportion indicates the degree of media coverage of a particular topic at a specific point in time. However, the topic proportions do not indicate whether the tone of the topic is positive or negative. We, therefore, also use tone-adjusted (sentiment) topic proportions. To investigate whether textual data can be used to improve tail risk forecasts, we use three different sets of predictors: (i) macroeconomic predictors only, (ii) macroeconomic predictors *plus* unadjusted text-based predictors, and (iii) economic predictors *plus* unadjusted text-based predictors *plus* text-based predictors with tone adjustment.[1]

In terms of forecasting models, two key insights guided our choice. First, mechanisms to prevent overfitting are necessary for OOS forecasting in high-dimensional settings like ours. We, therefore, use three Bayesian quantile regressions (QRs) with different shrinkage priors as Carriero, Clark, and Marcellino (2022b) have shown the importance of using shrinkage for QR in empirical macroeconomics. Second, recent studies point to the importance of capturing non-linear predictive relationships. For example, Goulet Coulombe, Leroux, Stevanovic, and Surprenant (2022) consider capturing non-linearities as the "true game changer" of machine learning methods for macroeconomic forecasting. This is corroborated by Medeiros, Vasconcelos, Veiga, and Zilberman (2021) and Clark et al. (2023), who find strong empirical support for tree-based methods. We use non-parametric Bayesian Gaussian Process Regressions (Williams & Rasmussen, 2006) and QR Forests (Meinshausen, 2006) to capture non-linear relationships between the quantile-specific target variables and the covariates. Entertaining both forecasting methods that feature linear and non-linear predictive relationships enables us to evaluate the empirical differences between the model classes.

While most studies analyze the benefits of text-based predictors for macroeconomic point forecasts, the role of text-based predictors for tail forecasts is largely unexplored. A few exceptions are Barbaglia, Consoli, and Manzan (2023), Sharpe, Sinha, and Hollrah (2023), and Filippou, Mitchell, and Nguyen (2023) who use sentiment-based approaches to determine the relevance of textual data (newspaper articles and FOMC speeches) for macroeconomic quantile forecasts. Our approach differs from these studies in three important ways. First, we investigate the added value (if any) of textual data in a setting where the forecaster is working in a data-rich environment using the FRED dataset, which contains a large amount of information about the economy. In this setting, we ask whether a large set of text-based predictors can provide additional value for tail risk predictions. Second, we consider forecasting models that can deal with high-dimensional predictor sets and two methods that can deal with non-linear predictive relationships. Third, we combine unsupervised topic models to extract the content of newspaper articles via estimated topic proportions and further tone-adjust these topic proportions by multiplying them with sentiment scores based on a pre-defined dictionary. By using both unadjusted and tone-adjusted text-based predictors, we can disentangle the added value of textual predictors that embed only information about media coverage from textual data, including information about the tone of newspaper articles. Our approach to learning from the news data is agnostic about which themes are relevant for predicting quantiles of macroeconomic times series. This contrasts with approaches such as that of Barbaglia et al. (2023), who only use text in an article that is semantically dependent on a term of interest (e.g., inflation).

Our key findings can be summarized as follows. First, textual data contain valuable predictive information not embedded in the large set of FRED-MD economic indicators. Our results suggest that text-based predictors can improve tail risk forecasts of macroeconomic variables but much less for the center of their probability distributions, consistent with the narrative that news signals are most helpful in extreme economic situations. Second, adding tone-adjusted text-based predictors results in more accurate forecast results than using only unadjusted predictors. Third, methods that can capture non-linear predictive relationships produce better predictions than those that cannot. We have conducted several robustness checks that confirm our main findings. Among other things, our results are robust to longer forecast horizons and a range of choices to produce text-based predictors.

[1] Text-based predictors are a viable alternative to survey forecasts and other "soft" data for quantile predictions. While survey forecasts were valuable in improving point forecasts of macroeconomic time series, they were not found to help produce more accurate variance estimates. The reason is that professional forecasters tend to be overconfident, hence providing too narrow density forecasts; see, e.g., Banbura, Brenna, Paredes, and Ravazzolo (2021), Galvão, Garratt, and Mitchell (2021). Given our focus on quantiles, particularly the tails of the distribution, textual data appear more promising. The available data history is considerably shorter than our text-based predictors for other "soft" data, such as Google Trends.

The paper proceeds as follows. Section 2 lays out our methodology. Section 3 outlines our forecasting setup, introduces our predictors, and presents our forecast results. Section 4 concludes. Additional material is relegated to the appendix.

2. Methodology

2.1. Forecasting methods

2.1.1. Bayesian quantile regressions

In settings with many regressors, Bayesian shrinkage alleviates overfitting and thus noisy forecasts. Recently, Carriero et al. (2022b) showed the importance of using shrinkage for QR in empirical macroeconomics. For Bayesian estimation, we use the mixture representation established by Yu and Moyeed (2001). For a given variable y of interest that is to be predicted for quantile τ at horizon h, the Bayesian QR can be stated as

$$y_{t+h} = x_t'\beta_\tau + \varepsilon_{\tau,t+h}, \tag{1}$$

where $\{x_t\}_{t=1}^T$ is a column vector of predictors, β_τ denotes a vector of quantile-specific regression coefficients and $\varepsilon_{\tau,t+h}$ follows an asymmetric Laplace distribution, see Yu and Moyeed (2001). The dimension of x_t depends on the setting, which is one of the following three possibilities: (i) FRED predictors only, (ii) FRED predictors *plus* unadjusted text-based predictors, and (iii) FRED predictors *plus* unadjusted text-based predictors *plus* text-based predictors with tone adjustment.

Posterior inference requires specifying a likelihood and eliciting priors for the coefficients. As Markov Chain Monte Carlo estimation is slow in high dimensions, we use fast variational Bayes approximations for posterior inference. In the interest of brevity, we do not give full details of the estimation, but note that by introducing auxiliary latent variables, the likelihood is conditionally Gaussian, and the errors are conditionally heteroscedastic, see Yu and Moyeed (2001) and Prüser and Huber (2024). The shrinkage priors we consider fall into the class of global-local shrinkage priors, where the prior variance comprises one global term pertaining to all coefficients and another coefficient-specific term. Our chosen shrinkage priors can be written in the general form

$$\beta_\tau | \psi_{\tau_1}, \ldots, \psi_{\tau_J}, \lambda_\tau \sim \prod_{j=1}^{J} \mathcal{N}\left(0, \psi_{\tau j}\lambda_\tau\right), \quad \psi_{\tau j} \sim u, \quad \lambda_\tau \sim \pi, \tag{2}$$

where J denotes the number of predictors, λ_τ denotes a quantile-specific global shrinkage parameter and $\psi_{\tau j}$ are quantile-specific local scaling parameters that control the coefficient-specific shrinkage intensities. Different shrinkage priors are generated by choosing different mixing densities via the u and π functions. We consider three shrinkage priors that are popular choices in macroeconomic forecasting; see, e.g., Huber and Feldkircher (2019), Cross, Hou, and Poon (2020), and Prüser (2023):

- Ridge: The Ridge prior collapses to a purely global shrinkage prior since all local scaling parameters are set equal to 1. The global shrinkage parameter follows an inverse Gamma distribution. Formally, we thus have: $\psi_{\tau j} = 1 \; \forall \tau, j$ and $\lambda_\tau \sim \mathcal{IG}(e_0, e_1)$. We choose the hyperparameters $e_0 = e_1 = 0$ which results in a flat prior, see Tipping (2001). Overall, the Ridge prior offers a comparatively low degree of flexibility for variable-specific deviations from the global shrinkage pattern. This prior is supposed to work well when many predictors are relevant, consistent with a dense representation of the prediction problem.
- Horseshoe: In contrast to the Ridge prior, the Horseshoe prior (Carvalho, Polson, & Scott, 2010) offers variable-specific shrinkage. The Horseshoe sets both u and π to half-Cauchy distributions: $\sqrt{\psi_{\tau j}} \sim \mathcal{C}^+(0, 1)$ and $\sqrt{\lambda_\tau} \sim \mathcal{C}^+(0, 1)$. An advantage of the Horseshoe prior is that it does not require the researcher to elicit any tuning parameters. The Horseshoe prior has been shown to have excellent posterior contraction properties, for example, Ghosh, Tang, Ghosh, and Chakrabarti (2016). The Horseshoe prior spikes at zero and has fat tails. Hence, it is supposed to shrink small coefficients of unimportant predictors to zero, but (in relative terms) large coefficients of the informative predictors are not shrunk much. Accordingly, the Horseshoe prior should work well when only a small number from the pool of predictors is useful, consistent with a sparse representation of the prediction problem.
- Lasso: The Lasso prior is a special case of the Normal-Gamma prior of Brown and Griffin (2010). The Lasso involves setting u and π to Gamma distributions: $\psi_{\tau j} \sim \mathcal{G}(1, \lambda_\tau/2)$ and $\lambda_\tau \sim \mathcal{G}(c_0, d_0)$. We set the hyperparameters $c_0 = d_0 = 0$.

2.1.2. Gaussian process regressions

Gaussian Process Regression (Williams & Rasmussen, 2006) is a non-parametric method recently used for inflation forecasting by Clark, Huber, Koop, and Marcellino (2024). It elicits a process prior on the function $g_\tau(x_t)$:

$$g_\tau(x_t) \sim \mathcal{GP}\left(\mu_\tau(x_t), \mathcal{K}(x_t, x_t)\right), \tag{3}$$

where we set the mean function $\mu_\tau(x_t)$ to zero. The kernel function $\mathcal{K}\left(x_t, x_t'\right)$ describes the relationship between x_t and x_t, for $t, t = 1, \ldots, T$.

As x_t is observed at discrete points in time, $g_\tau = (g_\tau(x_1), \ldots, g_\tau(x_T))'$:

$$g_\tau \sim \mathcal{N}(0_T, K(w)), \tag{4}$$

where $K(w)$ refers to a $T \times T$-dimensional matrix with (t, t)-th element $\mathcal{K}(x_t, x_t)$.

The type of kernel determines the estimated function. We choose a squared exponential kernel

$$\mathcal{K}(x_t, x_t) = w_1 \times e^{-\frac{w_2}{2}\|x_t - x_t\|^2}, \tag{5}$$

where we follow Chaudhuri, Kakde, Sadek, Gonzalez, and Kong (2017) for setting the hyperparameters $w = (w_1, w_2)'$, which govern the smoothness of the function.

Above, we have outlined the function-space view of the GP regression. In the alternative weight-space view, which is convenient for estimation, the GP regression can be expressed as

$$y = Z\gamma_\tau + \varepsilon, \tag{6}$$

where y denotes the stacked dependent variables, Z represents the lower Cholesky factor of K, and $\gamma_\tau \sim \mathcal{N}(0_T, I_T)$.

2.1.3. Quantile regression forests

Our last forecasting method is a frequentist non-parametric method, namely QR Forests, an extension of Random Forests (Breiman, 2001) for conditional point estimation to conditional quantile estimation based on an ensemble of trees (Meinshausen, 2006). Random Forests and QR Forests capture non-linear predictive relationships and, primarily due to this feature, have been found to perform well in macroeconomic forecasting; (see, e.g., Clark et al., 2023; Medeiros et al., 2021).

Random Forests grow a large number of trees by using t observations:

$$(Y_i, X_i), i = 1, \ldots, t,$$

where Y is the variable of interest and X is a (possibly high-dimensional) predictor variable. For ease of notation, we drop time subscripts. For each tree and node, Random Forests use a random subset of predictors to split on.[2] The intuition of this random selection is to de-correlate the trees and thus decrease the variance of the forecasts. In Random Forests, the conditional mean prediction of Y, given $X = x$, is generated as the weighted sum over all observations:

$$\widehat{\mu}(x) = \sum_{i=1}^{t} w_i(x) Y_i, \tag{7}$$

where the weights $w_i(x)$ are computed over the collection of trees. In each tree, the conditional mean prediction is the simple average of all observations that fall into the same leaf when dropping down x; the remaining observations are neglected.

Meinshausen (2006) extended the Random Forests to QR Forests. The conditional distribution function of Y, given $X = x$, is

$$F(y|X = x) = P(Y \leq y|X = x) = \mathbb{E}\left(\mathbb{1}_{\{Y \leq y\}}|X = x\right). \tag{8}$$

Instead of approximating the conditional mean $\mathbb{E}(Y|X = x)$ in case of Random Forests, for QR Forests, $\mathbb{E}\left(\mathbb{1}_{\{Y \leq y\}}|X = x\right)$ is approximated by the weighted mean over the observations $\mathbb{1}_{\{Y_i \leq y\}}$,

$$\widehat{F}(y|X = x) = \sum_{i=1}^{t} w_i(x) \mathbb{1}_{\{Y_i \leq y\}}, \tag{9}$$

where $w_i(x)$ are the same weights as for Random Forests. Based on $\widehat{F}(y|X = x)$, we can estimate the desired conditional α-quantile $Q_\alpha(x)$ as

$$\widehat{Q}_\alpha(x) = \inf\left(y : \widehat{F}(y|X = x) \geq \alpha\right). \tag{10}$$

An important difference between QR Forests and Random Forests is that, for each node in each tree, Random Forests keep only the mean of the observations that fall in that node. In contrast, QR Forests keep the values of all the observations in that node to compute the conditional distribution.

2.2. Probabilistic topic models

We use a probabilistic topic model to create text-based predictors. Topic models are data-driven and use a probabilistic approach to identify themes within a large set of written documents. Unlike dictionary-based and Boolean approaches, topic models are initially context-agnostic, finding word clusters (topics) solely based on the co-occurrence of words.

The most prominent topic model is latent Dirichlet allocation (LDA) by Blei, Ng, and Jordan (2003). It posits that documents are generated by a stochastic process where each text is a mixture of latent topics. Each topic is a probability distribution over the same vocabulary, but with different probabilities for each word. Despite its prominence and advantages, Latent Dirichlet Allocation (LDA) cannot account for the fact that certain topics tend to co-occur within documents (e.g., inflation and commodity prices). We therefore use the more sophisticated CTM by Blei and Lafferty (2007), which was developed to address this limitation.[3]

Fig. 1 shows a graphical model of the generative process assumed by the CTM, where edges denote dependencies, nodes are random variables, and plates are replicated variables. The only observable variables of the model are the words (w).

The key difference between LDA and the CTM is the assumption for the topic proportions, θ_d. The CTM assumes that topic proportions follow a logistic normal distribution— with a mean vector (μ) and a covariance matrix (Σ)— instead of LDA, which assumes that topic proportions come from a Dirichlet distribution. Consistent with the more realistic assumptions of correlating topics, the CTM yields higher values for statistics that measure the predictive performance regarding unseen

[2] In our empirical work, for a p-dimensional predictive variable, at each node, we use the default choice of $\sqrt{p}$ randomly selected predictors.

[3] Economic studies have predominantly relied on LDA to extract news-based predictors. These predictors have then been used, for instance, to investigate the value of news data for modeling macroeconomic dynamics (Bybee et al., 2024; Larsen & Thorsrud, 2019), to construct a daily business cycle index (Thorsrud, 2020), to predict US macroeconomic variables (Ellingsen et al., 2022), and to nowcast US GDP (Babii, Ghysels, & Striaukas, 2022) as well as Chinese GDP and inflation expectations (Zheng, Fan, Jin, & Fang, 2024). The CTM and further extensions have been used to produce topic proportions which have then been used, for example, to analyze news coverage of China (Roberts, Stewart, & Airoldi, 2016), to investigate the impact of presidential tax speeches on economic activity (Dybowski & Adämmer, 2018), to forecast the equity premium (Adämmer & Schüssler, 2020), and to analyze European Central Bank communication (Bohl, Kanelis, & Siklos, 2023; Dybowski & Kempa, 2020).

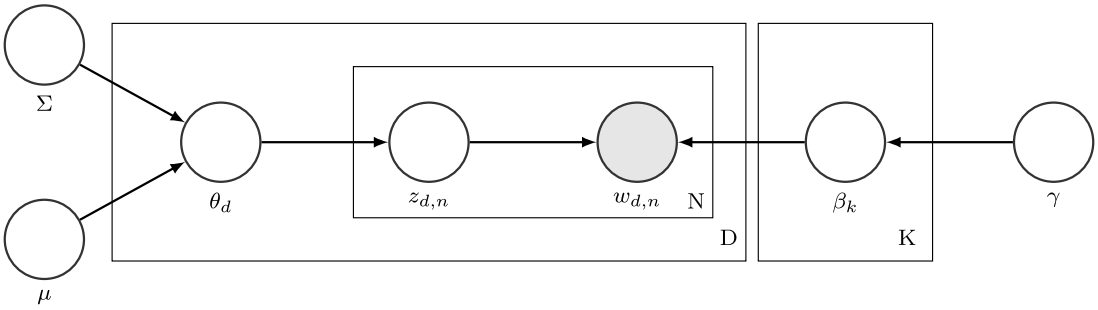


Fig. 1. Generative process of the CTM based on Blei and Lafferty (2007).

documents (Blei & Lafferty, 2007). The generative process of the CTM can be written as follows (local mathematical notation):

1. The probabilities of words in topic k are given in vector β_k. Each topic k is a multinomial distribution over a unique vocabulary, generated from the Dirichlet distribution with input vector γ:

   $\beta_k \sim Dir(\gamma)$.

2. For each document d and topic k, topic proportions are drawn from the logistic normal distribution:

   $$\eta_d \sim \mathcal{N}(\mu, \Sigma)$$
   $$\theta_{d,k} = \frac{\exp(\eta_{d,k})}{\sum_{i=1}^{K} \exp(\eta_{d,i})},$$

   where K denotes the number of topics. For each document d, we obtain the K-dimensional vector of topic proportions as $\theta_d = (\theta_{d,1}, \ldots, \theta_{d,K})'$.

   (a) For each word n, a topic assignment is drawn from the multinomial distribution:

   $z_{d,n}|\theta_d \sim Mult(\theta_d)$.

   (b) Each word is drawn from the multinomial distribution:

   $w_{d,n}|z_{d,n}, \beta_{1:K} \sim Mult(\beta_{z_{d,n}})$.

We use the *partially collapsed variational EM algorithm* by Roberts et al. (2016) to estimate the CTM. The approach is implemented in the R-package `stm` by Roberts, Stewart, and Tingley (2019). We are particularly interested in θ_d, namely the topic proportions of each document, whose aggregates serve as our text-based predictors; see Section 3.2.2.

3. Empirical work

3.1. Forecasting setup

We generate monthly quantile forecasts (nowcasts and one month ahead) for employment, inflation (Total CPI), industrial production, and consumer sentiment. We choose employment, inflation, and industrial production as our target variables because they are key macroeconomic variables used in several forecasting studies; for example, Barbaglia et al. (2023), Kalamara et al. (2022). Consumer sentiment assesses whether news data are important in forming household expectations (Larsen, Thorsrud, & Zhulanova, 2021). Depending on the setting, we incorporate FRED predictors, text-based predictors, or both together. We detail our predictors in Section 3.2. In addition, all model specifications include 12 lags of the respective (transformed) variable of interest.

We use vintage data from the McCracken and Ng (2016) database for our target variables and the FRED predictors. We use 98 indicators from the FRED-MD database and transform the data to achieve stationarity (McCracken & Ng, 2016).[4] Our estimation sample starts in 1980:06 when the text-based predictors start. We run recursive estimations based on an expanding window. Our evaluation period runs from 1999:08 to 2021:12.

We compute the predictions at the end of a given month. Due to publication lags, we include macroeconomic predictors from the previous month and text-based and financial predictors from the said month.[5] For example, if we are at the end of December and produce a nowcast for December, we use the macroeconomic predictors from November released in December and the financial and text-based predictors from December. Similarly, for one-month-ahead forecasts, if we are at the end of December and produce a prediction for January, we use the macroeconomic predictors from November released in December and the financial and text-based predictors from December.

Our five forecasting models are three versions of Bayesian QRs with different shrinkage priors (Horseshoe, Ridge, and Lasso), QR Forests, and Gaussian Process Regressions.

We evaluate our forecasting models with the quantile score (QS), which is computed as

$$QS_{\tau,t+h} = \left(y_{t+h} - Q_{\tau,t+h}\right)\left(\tau - \mathbb{1}_{\{y_{t+h} \leq Q_{\tau,t+h}\}}\right), \tag{11}$$

where y_{t+h} is the actual outcome of the variable of interest in $t+h$, $Q_{\tau,t+h}$ denotes the forecast of quantile τ for $t+h$. The indicator function $\mathbb{1}_{\{y_{t+h} \leq Q_{\tau,t+h}\}}$ takes on a

[4] The transformation of the variables can be found in Appendix G. We use all indicators from the FRED-MD database that are available at the beginning and end of our out-of-sample period.

[5] See Appendix G for which variables are classified as financial predictors. Note that a few indicators are lagged by more than one month, and we adjusted for this.

value of 1 if the outcome is not higher than the quantile forecast, and 0 otherwise.

3.2. Predictor sets

3.2.1. Economic predictors

We select all macroeconomic predictors from the McCracken and Ng (2016) FRED-MD database that are available at both the end and the beginning of the sample. The predictors can be classified into eight categories: (i) output and income, (ii) labor market, (iii) housing, (iv) consumption, orders, and inventories, (v) money and credit, (vi) interest and exchange rates, (vii) prices, and (viii) stock market. For the classification of the variables, see https://www.ssc.wisc.edu/~bhansen/econometrics/FRED-MD_description.pdf.

3.2.2. Text-based predictors

We used the database NexisUni from LexisNexis to download 793,013 economically related newspaper articles from *The New York Times* and *The Washington Post* between 1980:06 and 2021:12. We then conducted Part-of-Speech-Tagging (Benoit & Matsuo, 2022) to remove everything but lemmatized nouns from the articles. Our reason for this choice is that topic models aim to summarize the *content* of documents, which is predominantly described by nouns, in contrast to sentiment analysis, which aims to describe the documents' *tone*. In addition, Martin and Johnson (2015) found the highest values of semantic coherence when using a nouns-only approach, a metric that strongly correlates with human judgment.

We constructed a document-term matrix (dtm) based on the tokenized nouns. A dtm counts how often a word (column) occurs within a document (row). We then computed term-frequency inverse-document-frequency (tf-idf) values for each word to extract the 10,000 most relevant terms until 1999:08. The final dtm served as the input for the CTM. In addition, the researcher has to give the number of topics K as input. Ellingsen et al. (2022), Larsen and Thorsrud (2019), Thorsrud (2020) use LDA to compute topic proportions for different inference and prediction tasks. In each study, they choose 80 topics. We follow them and use $K = 80$ as our baseline model.[6]

We estimated for each document on each day 80 topic proportions based on the newspaper articles until 1999:08. After that, we computed the topic proportions OOS to exclude any look-ahead bias, similar to Ellingsen et al. (2022). Finally, we computed the simple average of all the estimated topic proportions of the documents for a given month. Note that $\theta_{d,k}$ denotes the (estimated) topic proportion of the kth topic for the dth document. For a given month m and for each topic k, we computed the average topic proportion as follows:

$$\theta_k^{(m)} = \frac{1}{D_m} \sum_{d=1}^{D_m} \theta_{d,k}, \tag{12}$$

where D_m denotes the number of documents that fall into the mth month. Applying (12) to all K (estimated) topic proportions produces a $K \times 1$ vector of (estimated) topic proportions for the mth month, that is $\theta^{(m)} = (\theta_1^{(m)}, \ldots, \theta_K^{(m)})'$.

Topic proportions only reveal information about the content of the newspaper articles, not about the tone accompanying them. However, the tone may contain additional predictive information. We, therefore, also consider tone-adjusted (sentiment) topic proportions as predictors. To do so, we rely on the dictionary compiled by Barbaglia et al. (2023), where each word contains a sentiment value between -1 and 1. The more positive (negative) the sentiment score, the more positive (negative) the tone of the respective word. Unlike the nouns-only approach for calculating the topic proportions, we use all the words in the sentiment dictionary to calculate a document's sentiment score. The sentiment score (*Sent*) for the dth document is computed as follows:

$$Sent_d = \frac{\sum_{i=1}^{N} Score_i}{\#\text{Sentiment Words in } d}, \tag{13}$$

where $Score_i$ is the sentiment score for the ith word in the respective document, and the number of words in each article is denoted by N. The sum of a document's sentiment score is normalized by the number of words with an associated sentiment score in the respective text. Fig. 2 shows our monthly sentiment time series, computed as $Sent^{(m)} = \frac{1}{D_m} \sum_{d=1}^{D_m} Sent_d$. Gray-shaded areas indicate NBER-dated recessions. The time series captures economic periods of booms and busts well.

We use the sentiment score of each document to adjust each topic proportion:[7]

$$\text{SentTopic}_{d,k} = \theta_{d,k} \times Sent_d. \tag{14}$$

We then compute the average over all documents of a given month m:

$$\text{SentTopic}_k^{(m)} = \frac{1}{D_m} \sum_{d=1}^{D_m} \text{SentTopic}_{d,k}. \tag{15}$$

The first row of Fig. 3 shows the trajectories of three selected topics without any tone adjustment. A high value for a certain topic at a given point in time indicates high media coverage of said topic at that point in time.[8] The figures show that our topics capture important political and economic events such as several debt crises in Mexico, Asia, and Europe (Topic 9), the beginning of the Gulf

[6] In Appendix B, we provide robustness checks for the number of topics, which are based on topic interpretability (Adämmer & Schüssler, 2020; Roberts et al., 2019) and an automated search approach by Mimno and Lee (2014). Overall, our main findings are robust to alternative choices of the number of topics.

[7] Ellingsen et al. (2022) also tone-adjust their topic proportions, but in a different manner. In addition, they use the *Harvard IV-4 Psychological Dictionary*, which only distinguishes between positive and negative words and was not designed for economic and financial applications.

[8] The labels are based on the top five words of each topic. We emphasize that topic labeling is notoriously subjective, so we show all topics' top five words in Appendix A.

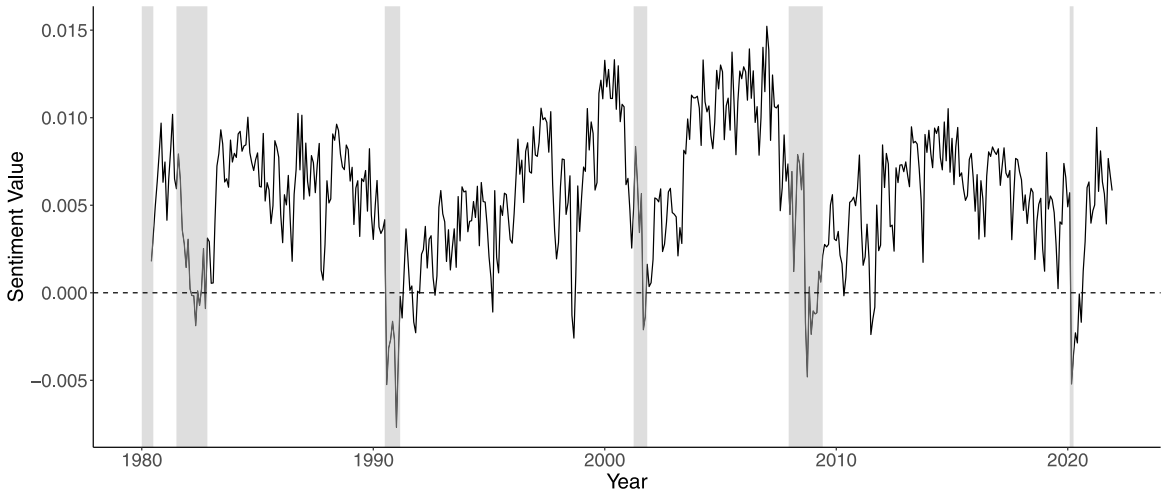


Fig. 2. Monthly sentiment scores based on the dictionary compiled by Barbaglia et al. (2023). Gray-shaded areas indicate NBER-dated recessions.

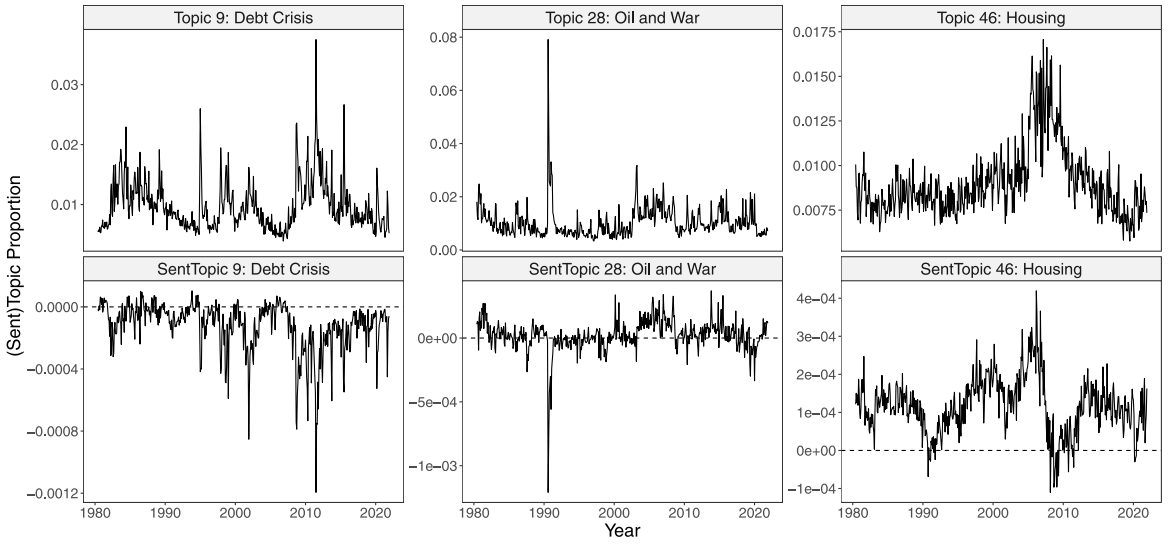


Fig. 3. Selected monthly unadjusted topic proportions (first row) and with tone-adjustment (second row) based on the dictionary of Barbaglia et al. (2023).

War in 1991 (Topic 28), and the Great Recession, which emanated in the housing market (Topic 46).

The second row of Fig. 3 shows the tone-adjusted topic proportions, reflecting several events' severity. For example, the tone of articles about debt crises (Topic 9) was negative overall. The series of SentTopic 28 drops sharply at the beginning of the Gulf War in 1991, and SentTopic 46, which mainly covers issues related to the housing market, clearly changes in sentiment from positive to negative as the collapse of the US housing market became apparent during the Great Financial Crisis. We use both the monthly averages of unadjusted topic proportions (12) and tone adjustment (15) as text-based predictors in our empirical analyses.

3.3. Forecast results

We first focus on the results for the linear Bayesian QR models with different shrinkage priors, namely Horseshoe, Lasso, and Ridge. Our primary interest here is to assess the text-based predictors' added value (if any). We then assess how methods that use the same sets of predictors but allow for non-linear predictive relationships (QR Forests and Gaussian processes) perform against the overall most successful linear model.

For Horseshoe, Lasso, and Ridge, Fig. 4 summarizes the results for nowcasts and one-month ahead forecasts, respectively. Each column denotes a target variable, and each row is a (linear) model. Forecast accuracy is assessed

by quantile scores across different quantiles τ: τ = 5%, 10%, 25%, 50%, 75%, 90%, 95%. A quantile AR(1) model serves as our benchmark.[9] Relative quantile scores below (above) one indicate more (less) precise forecasts compared to the benchmark model.

The combination of macroeconomic and text-based predictors leads to accuracy gains in many cases, especially in the tails, compared to the setting with macroeconomic predictors alone. The addition of tone-adjusted topic proportions leads to even lower overall quantile scores compared to economic predictors and unadjusted topic proportions (FRED & Topics). In those cases where the addition of text-based predictors does not improve forecast performance, it does not have a materially adverse effect either. The incremental value from adding text-based predictors is pronounced in the tails. Improved tail forecasting with textual data suggests a scenario where text-based predictors are particularly helpful in extreme economic environments, where timely news data can reflect the overall economic picture and be useful predictors of macroeconomic variables. In times such as the Great Recession or the COVID-19 pandemic, uncertainty, and political change may have been better captured than by hard economic indicators.

Within the class of the three linear models, the Horseshoe shrinkage prior underperforms the Lasso and Ridge in the tails. Ridge, as a purely global shrinkage prior, outperforms Horseshoe and Lasso. Thus, the richer shrinkage patterns offered by Horseshoe do not pay off in our analysis. These results are consistent with a dense representation of the prediction problem, where many weak predictors are relevant, rather than a sparse structure, where a few important predictors can be selected and the others are discarded.

Ridge produces more precise nowcasts for the left tails of employment and production and for both tails of inflation and consumer sentiment relative to a quantile AR(1) model when using FRED & Topics & SentTopics. When using FRED & Topics & SentTopics, Ridge is also successful in one month ahead forecasting for the left-tail of production and the right-tail of inflation and employment. For consumer sentiment (nowcasts and one-month ahead forecasts), quantile scores markedly improve in the left tail when (in particular tone-adjusted) text-based predictors are added. This result is interesting from an information processing aspect, supporting that news data play an important role for households in forming their expectations (Larsen et al., 2021).

Next, we look at the performance of non-linear methods, namely Gaussian Processes and QR Forests. We are interested in the added value of non-linear models over linear ones, so we use Ridge with the full predictor set (FRED & Topics & SentTopics) as our benchmark. We do this because Ridge was overall the most successful linear model, particularly in the tails (see Fig. 4). Fig. 5 shows the relative predictive performance of both non-linear methods with the various predictor sets. Relative quantile scores below (above) one indicate more (less) precise forecasts compared to Ridge with all predictors (FRED & Topics & SentTopics).

As indicated by the predominantly hump-shaped relative quantile scores, with many values below one in Fig. 5, both non-linear methods were overall better at tail forecasting than Ridge. This is particularly true for the right-tail outcomes of employment and both tails of production, especially when using the FRED & Topics & SentTopics predictor set. Hence, combining non-linear methods and a mix of economic and text-based predictors seems promising for macroeconomic tail forecasting. That said, the marginal difference in quantile scores from adding text-based predictors is smaller overall for non-linear methods than for linear ones. One explanation for this finding may be that text-based predictors may partially compensate for missing non-linear predictive relationships in the linear models.

3.4. Robustness checks and further analyses

We conducted a number of robustness checks and further analyses to explore how alternative settings affect our findings in this paper.

Generating text-based predictors. The computed topic proportions obtained from the CTM are influenced by a number of choices. Here, we explore the impact of alternative choices and examine their potential impact on forecast accuracy. First, we computed $K = 68$ topics based on the random search approach by Mimno and Lee (2014). More specifically, we ran their algorithm 100 times and chose the average estimated number of topics across all runs, 68. Second, following Adämmer and Schüssler (2020), who use the CTM and a similar corpus, we computed $K = 100$ instead of $K = 80$ topics. The authors show empirically that $K = 100$ strikes a good balance between predicting unseen documents and topic interpretation. Third, the CTM is nested within the more sophisticated structural topic model (STM) by Roberts et al. (2016), which further allows the inclusion of covariates. We computed an STM to account for the two different newspapers. Fourth, we estimated our baseline model with 15,000 unique words instead of 10,000 unique words based on the highest tf-idf scores. Fifth, for our reported results, we used a training sample for the CTM from 1980:06 to 1999:08 (see also Section 3.1) and computed OOS topic proportions for the evaluation period from 1999:09 onwards. This approach avoids look-ahead bias but implicitly assumes that the vocabulary remains constant. In reality, however, this was certainly not the case. For example, words like "Brexit" or "COVID" were not part of the vocabulary in the estimation sample of the topic proportions. To investigate whether—and if so, how much—a fixed vocabulary lowers forecast accuracy, we estimated the CTM using the vocabulary of the whole sample (i.e., with a look-ahead bias). Interestingly, we did not get lower quantile scores overall. These results are consistent with those of van Dijk and de Winter (2023),

[9] More precisely, we compute a (frequentist) quantile AR(1) model. For example, in the case of a nowcast at the end of December (for December), the target variable's latest available value from November is used as the single predictor in the quantile AR(1): $y_{December} = \beta_{0,\tau} + \beta_{1,\tau} y_{November} + \varepsilon_{December}$.

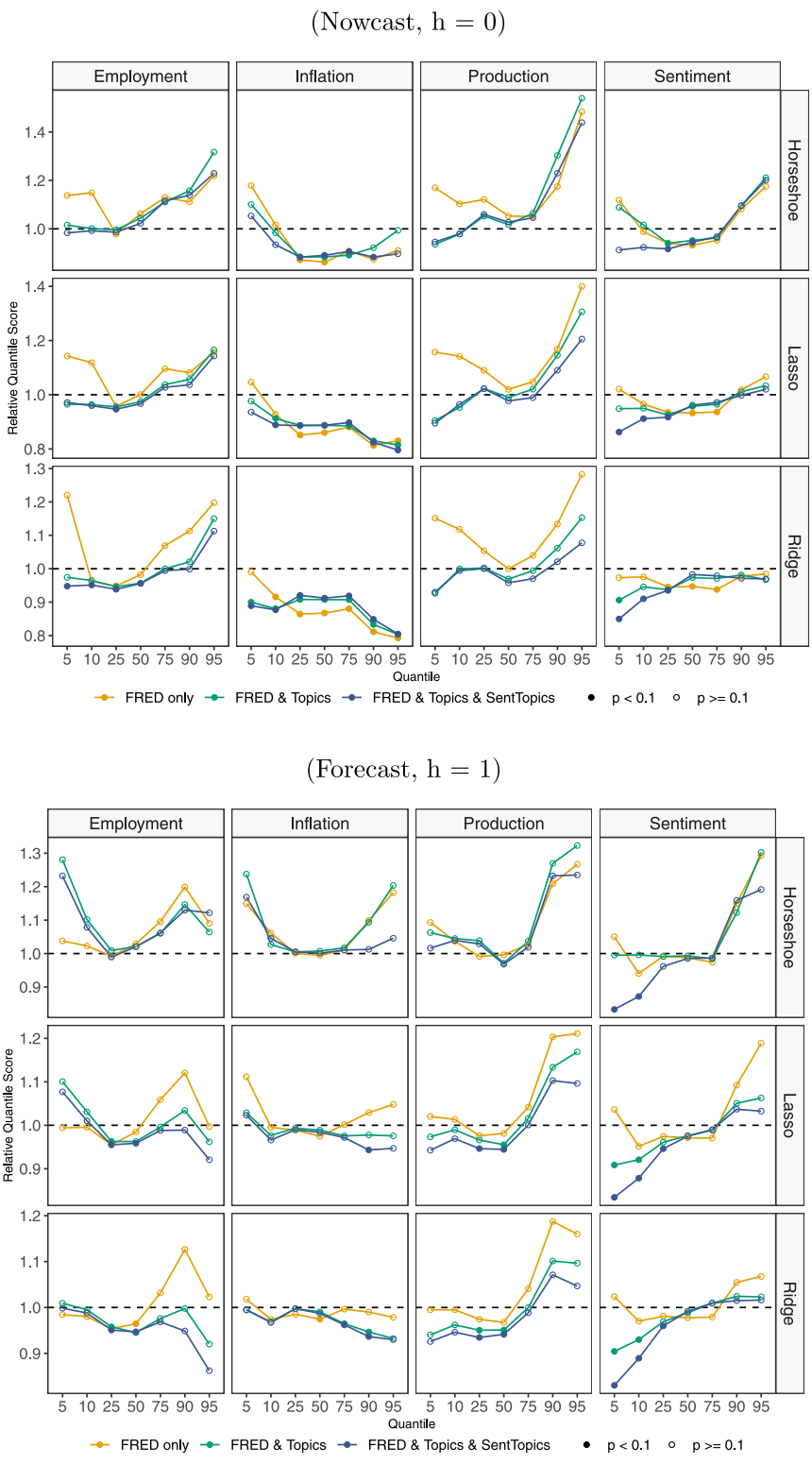


Fig. 4. Evaluation of quantile nowcasts (h = 0) and quantile one-month ahead forecasts (h = 1) for the Bayesian QRs with different shrinkage priors (Horseshoe, Ridge, and Lasso). The dotted black horizontal line shows the quantile score of the quantile AR(1) benchmark, which is standardized to 1.0. Scores below (above) 1.0 indicate more (less) precise forecasts for a given quantile compared to the quantile AR(1) benchmark. Inside colored dots indicate significantly higher forecast accuracy compared to the quantile AR(1) benchmark according to a one-tailed Diebold and Mariano (1995) test at the 10% level.

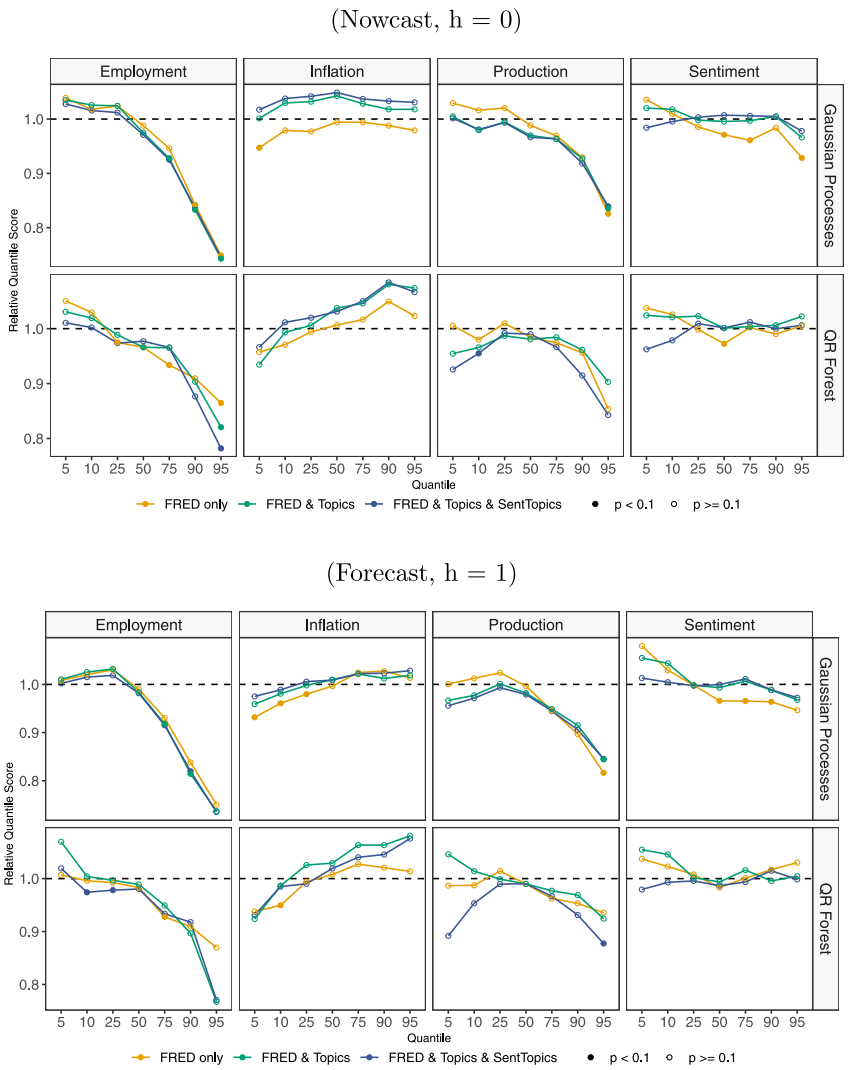


Fig. 5. Evaluation of quantile nowcasts (h = 0) and quantile one month ahead forecasts (h = 1) for Gaussian Processes and QR Forests. The dotted black horizontal line shows the quantile score of Ridge with the full predictor set FRED & Topics & SentTopics, which is standardized to 1.0. Scores below (above) 1.0 indicate more (less) precise forecasts for a given quantile than the Ridge benchmark. Inside colored dots indicate significantly higher forecast accuracy compared to Ridge according to a one-tailed Diebold and Mariano (1995) test at the 10% level.

who accounted for shifts in the vocabulary used to nowcast GDP but did not find gains in forecast accuracy either. Sixth, we computed separate topic proportions for the last week of the month and for each topic. However, based on more recent news, we found no added value from this alternative specification. Overall, the robustness checks reveal that the empirical results are not sensitive to the exact specification used to generate the topic proportions. Figure 8 and Figure 9 in Appendix B summarize the results for the different choices. The predictor set used in each setting is FRED & Topics & SentTopics. We also added the results for our baseline model ($K = 80$) for comparison. Overall, the results are robust across different choices. In this paper we have taken a conservative approach by not fine-tuning the parameters that control the generation of text-based data. More sophisticated procedures for selecting these parameters may further increase the gains in macroeconomic quantile prediction but with an increased risk of overfitting.

Alternative forecast horizons. Figures 10 and 11 in Appendix C summarize the results for the 6-month and the 12-month forecast horizon. The results confirm our findings for the nowcast and one-month-ahead forecasts. The marginal improvement from adding (tone-adjusted) text-based predictors seems even higher at these horizons, especially for the non-linear models and tail predictions associated with economic downside risk (low employment, high inflation, and low production).

Quantile scores over time. Figure 16 and 17 in Appendix E show the cumulative quantile scores at the 10% quantile relative to the quantile AR(1) benchmark for the nowcasts and one month ahead forecasts, respectively; Figure 18 and 19 in Appendix E do the same for the 90% quantile. Note that the presentation of results starts

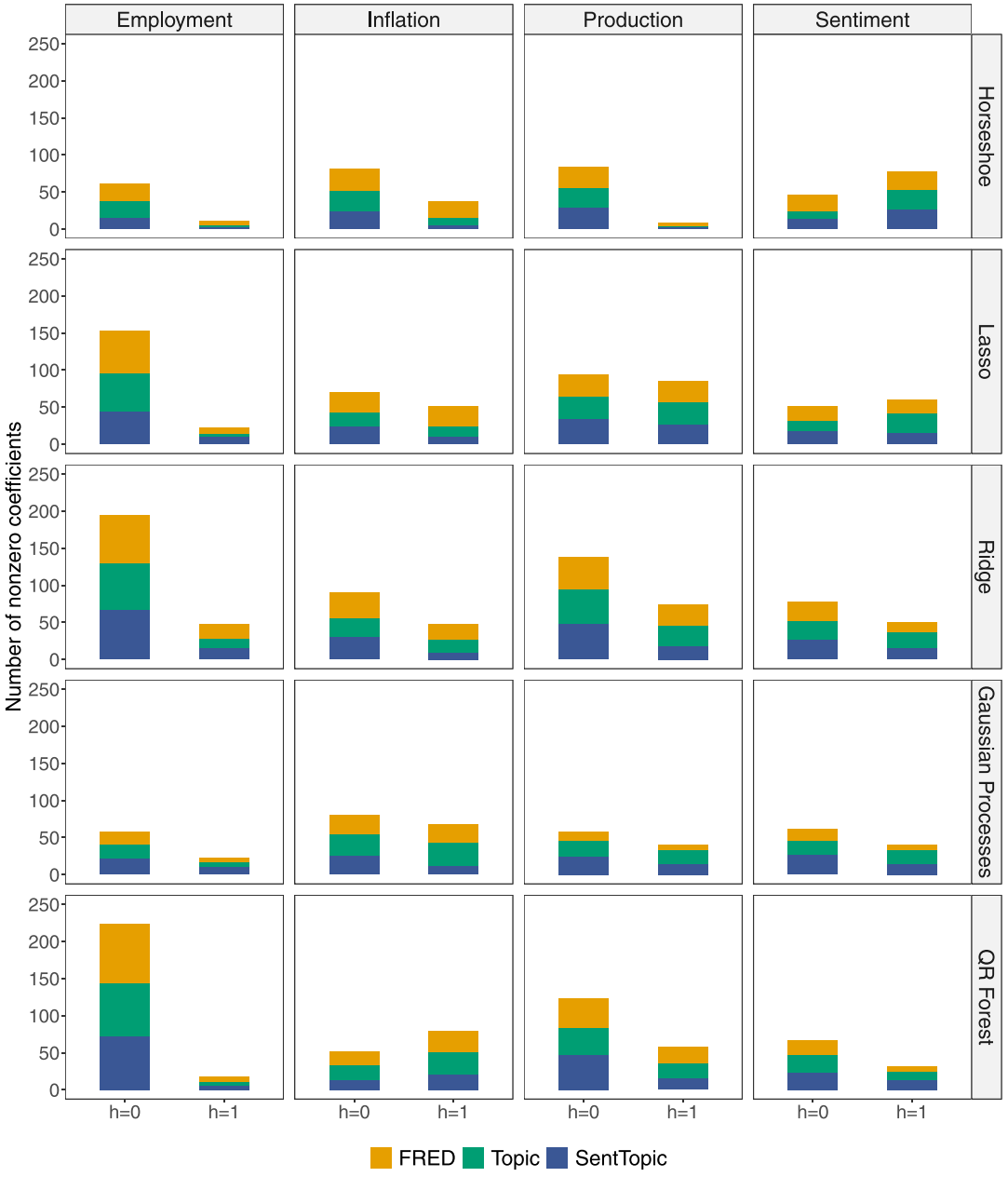


Fig. 6. Number of non-zero predictors in the regularized regression for the 10%-quantile ($\tau = 0.1$) for nowcasts (h = 0) and one month ahead forecasts (h = 1).

at 2005:01 rather than 1999:08, as earlier starting dates would distort the scaling due to the few observations at the beginning of the OOS period. Overall, across the different target variables and methods, gains relative to the quantile AR(1) benchmark are most pronounced during the Great Recession and the COVID-19 pandemic. Thus, our results are consistent with the typical finding in the economic forecasting literature that the performance of forecasting models is unstable over time; see, for example, Rossi (2013). To assess the marginal gains of the text-based predictors, Figure 20 and 21 in Appendix F show the cumulative quantile scores at the 10% quantile

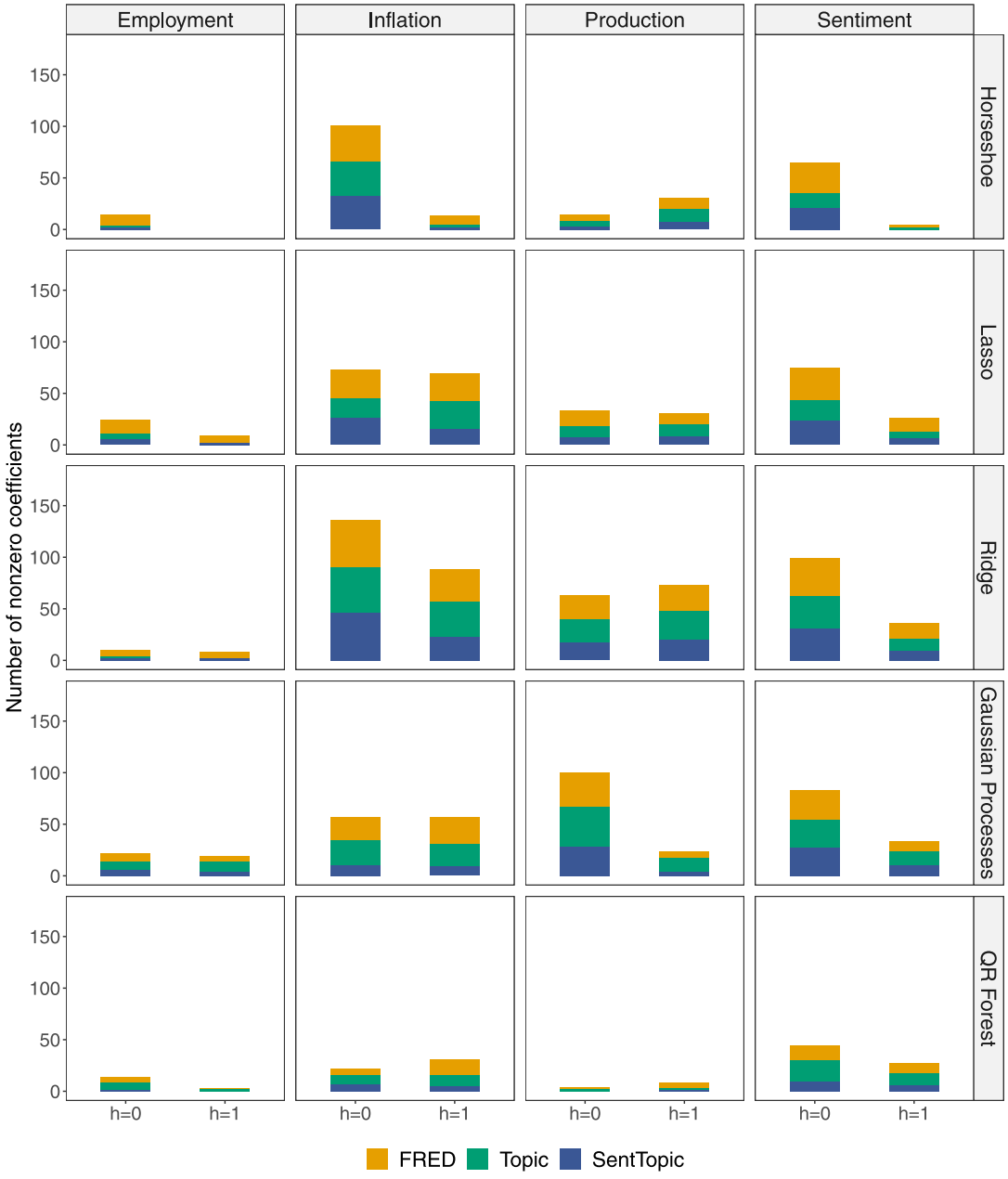


Fig. 7. Number of non-zero predictors in the regularized regression for the 90%-quantile ($\tau = 0.9$) for nowcasts (h = 0) and one month ahead forecasts (h = 1).

relative to the FRED-only dataset as the benchmark for the nowcasts and one month ahead forecasts, respectively; Figure 22 and 23 in Appendix F do the same for the 90% quantile. Across the different target variables and methods, we observe heterogeneous patterns and find that text-based predictors (both unadjusted and tone-adjusted) add value relative to the FRED-only dataset at different points in time rather than being restricted to short episodes.

3.5. Which predictors determine the quantile predictions?

As we include forecasting methods that feature non-linear predictive relationships, it is not obvious how to determine the marginal effect of a predictor on the variable of interest. Furthermore, even though we use heterogeneous forecasting methods, we want to ensure comparability of measures of predictor importance across methods. To accomplish this task and to shed light on the

predictors with the highest impact for each forecasting model, we rely on a linear approximation to the prediction distribution for each prediction model, as proposed by Woody, Carvalho, and Murray (2021) and used by Clark et al. (2024) and Prüser and Huber (2024). Concretely, we approximate the quantile predictions $Q_{\tau,t+h}$ with a linear regression model that uses regularization. For each quantile τ, the following Lasso-type optimization problem is solved:

$$\beta_\tau^* = \arg\min_{\beta_\tau} \sum_{t=t_0}^{T-h} \left(Q_{\tau,t+h} - \beta_\tau' x_t\right)^2 + \lambda \sum_{j=1}^{J} \left|\beta_{\tau,j}\right|, \qquad (16)$$

where t_0 denotes the beginning of the hold-out period, $\beta_\tau = \left(\beta_{\tau,1}, \ldots, \beta_{\tau,J}\right)'$ denotes the vector of coefficients, J denotes the number of potential predictors, and $\lambda \geq 0$ is a penalty parameter that controls the shrinkage intensity and is chosen via cross-validation. We use our most comprehensive set of predictor variables for the predictor importance analysis, including both FRED and all text-based predictors. We focus on the 10% and 90% quantiles because our previous analysis has shown the most pronounced patterns in the tails.

Figs. 6 and 7 show the number of nonzero coefficients for the nowcasts and one month ahead forecasts of the 10% and the 90% quantile, respectively. Across different target variables and quantiles, the structure of included FRED and text-based predictors is balanced. Within text-based predictors, roughly the same number of tone-adjusted and unadjusted topic proportions are included.

Figure 12 and 13 in Appendix D report the five most influential predictors for nowcasts and one month ahead forecasts at the 10%-quantile, respectively. Figure 14 and 15 in Appendix D do the same for the 90%-quantile. Importance is measured by the absolute values of the coefficients associated with the (standardized) predictors.

Overall, the coefficients not set to zero are of different types: lagged values of the respective target variable, FRED economic indicators, unadjusted topic proportions, and tone-adjusted topic proportions. Regarding the text-based predictors, the chosen topic proportions make narrative sense for the respective target variable. For instance, Topic 46 and SentTopic 46 were chosen for inflation nowcasts and one-month ahead forecasts. The five most probable words in that topic are related to the housing market (*home*, *house*, *property*, *estate*, and *owner*). Developments in the housing market can be related to inflation dynamics. Topic 31, related to household finance, was also chosen for inflation nowcasting by QR Forests. For nowcasts and one month ahead forecasts of employment, Topic 47 and SentTopic 47 were chosen by Gaussian Processes, which makes sense given the topic's job market-related words *job*, *work*, *force*, *unemployment* and *employment*. In addition, Topic 55 and SentTopics 55 were frequently selected for nowcasts and one-month ahead forecasts of employment and production. The topic is related to nutrition and health issues, given the topic's most prominent words *food*, *disease*, *meat*, *animal*, and *product*.

4. Concluding remarks

We have analyzed the added value of text-based predictors for quantile predictions of macroeconomic time series. Our high-dimensional setup included forecasting methods with both linear and non-linear quantile-specific predictive relationships, and we considered different sets of predictors.

Although forecast performance varied across quantiles and target variables, combinations of FRED and text-based predictors produced the most accurate forecasts, especially in the tails. Tone-adjusted text-based predictors added further value as predictors compared to unadjusted ones alone. Overall, Gaussian Process Regressions and QR Forests prevailed in terms of tail forecast accuracy, suggesting that non-linear predictive relationships are a promising route to follow in tail forecasting.

In summary, our empirical results suggest that researchers using FRED data for tail risk forecasting should exploit the additional information contained in textual data. The most precise tail risk predictions can be achieved using non-linear models with economic indicators and (tone-adjusted) topic proportions. Our results can be of value to researchers, forecasters, and policymakers alike.

CRediT authorship contribution statement

Philipp Adämmer: Conceptualization, Data curation, Funding acquisition, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **Jan Prüser:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Software, Validation, Writing – original draft, Writing – review & editing. **Rainer A. Schüssler:** Conceptualization, Funding acquisition, Methodology, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Rainer A. Schüssler reports financial support was provided by German Research Foundation. Philipp Adämmer reports financial support was provided by German Research Foundation. Jan Prüser reports financial support was provided by German Research Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2024.05.007.

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